
USAGE RELATEDNESS FOR ENGLISH SOCIAL MEDIA: TRANSFERRING THE DUREL FRAMEWORK TO TEMPOWiC

REPLICATION AND EXTENSION OF SCHLECHTWEG ET AL. [2018]

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ABSTRACT

Schlechtweg et al. [2018] propose DUREl, a framework for annotating *Diachronic Usage Relatedness*, and apply it to German historical text (DTA). Their study produced a test set of 1,320 use pairs for 22 target words. This report runs the same framework on English instead of German. The corpus is **TempoWiC** [Loureiro et al., 2022a], an English Twitter benchmark in which two tweets from different years are labelled for whether a target word carries the same meaning. TempoWiC only contains cross-period pairs, so we rebuild the missing within-period pairs by sampling from the same tweets. The result is 10,097 use pairs for 34 target words in DUREl’s three groups: EARLIER, LATER and COMPARE. Human annotators are replaced by relatedness models, from lexical overlap to a fine-tuned cross-encoder. On the TempoWiC task, zero-shot target-word cosine reaches 63.6 macro-F1 and supervised training does not improve on it (61.9 ± 1.4 for the cross-encoder), because the training and test splits share no target words. At the word level, COMPARE correlates with the human change rate at $\rho = -0.61$ ($p = 0.0001$), which supports the original claim that COMPARE measures the *degree* of change. Δ LATER does not transfer. On 2019–2021 Twitter data it mostly reflects pandemic vocabulary, which makes later usage more homogeneous rather than less. The code, the data pipeline and all result files are released with this report.

1 Introduction

Work on lexical semantic change is limited more by evaluation data than by models [Schlechtweg et al., 2018]. DUREl addresses this by moving *synchronic* polysemy annotation into a diachronic setting. Annotators never see a sense inventory. They only judge how related two *uses* of a word are, on a four-point scale, and change is read off the resulting means. The original study is German, historical (DTA, 1750–1800 vs. 1850–1900) and annotated by people.

This report asks one question: does that design still work in English, on social media, over a two-year gap instead of a century, with models doing the rating? We use TempoWiC [Loureiro et al., 2022a], which is built from tweets sampled one year apart and already carries human labels. Those labels give us something the German study did not have, namely an independent measurement of how much each word changed.

The report contributes three things: a DUREl-style dataset for English social media (10,097 use pairs for 34 target words in three groups, rebuilt from TempoWiC, §4); relatedness models used in place of annotators, evaluated on the original TempoWiC task and put through an agreement analysis in the shape of DUREl’s Table 3 (§6.1, §6.2); and a numerical test of both DUREl change measures against human judgements, which the German study could not run (§6.4).

2 Related work

Annotating meaning in context. Judging pairs of *uses* rather than assigning senses from an inventory goes back to synchronic work on graded polysemy. Brown [2008] asks annotators how fine-grained sense distinctions really are, and Erk et al. [2013] show that annotators can rate word meaning in context reliably on a graded scale, with average pairwise correlations between 0.55 and 0.62. DUREl [Schlechtweg et al., 2018] takes that idea into the diachronic setting, which is the framework this report applies. The same line of work later scaled up into Word Usage Graphs: DWUG [Schlechtweg et al., 2021] collects around 100,000 human proximity judgements in four languages and clusters uses into senses instead of reading change off group means. Our study sits between the two. We keep DUREl’s simple group means, but we obtain the judgements from models rather than from people.

Table 1: Resources for meaning change and meaning in context. The unit judged is what a single annotation decision covers.

Resource	Languages	Genre	Unit judged	Size
DUREl [Schlechtweg et al., 2018]	German	historical prose	use pair, 1–4	1,320 pairs
DWUG [Schlechtweg et al., 2021]	4 languages	mixed historical	use pair, 1–4	~100k judgements
SemEval-2020 T1 [Schlechtweg et al., 2020]	4 languages	historical corpora	word, binary + rank	137 words
WiC [Pilehvar and Camacho-Collados, 2019]	English	dictionary examples	use pair, binary	7,466 pairs
TempoWiC [Loureiro et al., 2022a]	English	tweets	use pair, binary	3,297 pairs
This report	English	tweets	use pair, 1–4 (model)	10,097 pairs

Table 2: The four-point relatedness scale used by DUREl, after Brown [2008]. Our models produce scores on this scale by the mapping in §4.5.

Score	Label	Reading
4	Identical	the two uses carry the same meaning
3	Closely related	the same meaning, used in different contexts
2	Distantly related	related meanings, e.g. literal vs. figurative
1	Unrelated	homonymy: no shared meaning
0	Cannot decide	rater abstains (excluded from the means)

Benchmarks. Two benchmark families are relevant. The first evaluates change at the level of the word: SemEval-2020 Task 1 [Schlechtweg et al., 2020] asks systems to decide whether a target word lost or gained a sense between two corpora in English, German, Latin and Swedish, with gold data built from the annotation procedure above. The second evaluates meaning at the level of the pair: WiC [Pilehvar and Camacho-Collados, 2019] asks whether a target word carries the same meaning in two contexts, and TempoWiC [Loureiro et al., 2022a] is its diachronic, social-media version, where the two contexts are tweets from different years. TempoWiC is the corpus we use. Its pair-level labels are what let us test DUREl’s word-level measures against an independent human measurement, which neither benchmark family does on its own.

Computational models of change. Early work compares static embedding spaces trained per period and aligns them [Hamilton et al., 2016]; Kutuzov et al. [2018] survey that generation of methods. Contextualised encoders changed the picture, since a word no longer has one vector per period but one per use. Giulianelli et al. [2020] cluster contextual representations of individual uses and derive change measures from how those clusters move, which is close in spirit to DUREl’s use pairs. Loureiro et al. [2022b] take the other route and pre-train Twitter encoders per time period (TimeLMs); those models provide the strongest baselines in the TempoWiC shared task. Our relatedness models belong to the contextual-encoder family, without the time-specific pre-training, for the reason given in §5.

Models as annotators. Using a model where an annotation study would use people is now common, and it is what makes a study of this size possible without an annotation budget. The practice has a known cost: a model’s judgements are correlated with its own training data and architecture, so a panel of models is not a panel of independent raters. §6.2 quantifies that cost on our data by putting the models through exactly the agreement analysis the German study ran on its five annotators. Table 1 places this report next to the resources it borrows from.

Position of this report. The contribution is not a new model. It is a transfer of an annotation framework to a new language, genre and time scale, plus the first numerical check of its two change measures against human judgements of change.

3 Background

DUREl. A *use* of a word is the sentence it occurs in. Uses are shown in pairs and rated 1 (*Unrelated*), 2 (*Distantly related*), 3 (*Closely related*) or 4 (*Identical*); 0 means “cannot decide”. Pairs come from three groups: EARLIER (both uses from t_1), LATER (both from t_2) and COMPARE (one from each). Two measures follow from the group means:

$$\Delta\text{LATER}(w) = \text{Mean}_l(w) - \text{Mean}_e(w) \quad (1)$$

$$\text{COMPARE}(w) = \text{Mean}_c(w) \quad (2)$$

A negative ΔLATER means *innovative* change: a new meaning appears, so later pairs are less related. A positive value means *reductive* change, where a meaning is lost. COMPARE measures the degree of change and does not distinguish the two types. The paper also proposes a polysemy-normalised variant, $\Delta\text{COMPARE}(w) = \text{Mean}_c(w) - \text{Mean}_e(w)$.

Table 2 gives the rating scale, and Figure 1 shows how the three groups are formed. The design has one property that matters for everything below: a rater never has to name a sense, and never has to know that the study is about change at all. All the diachronic information sits in which group a pair was drawn from.

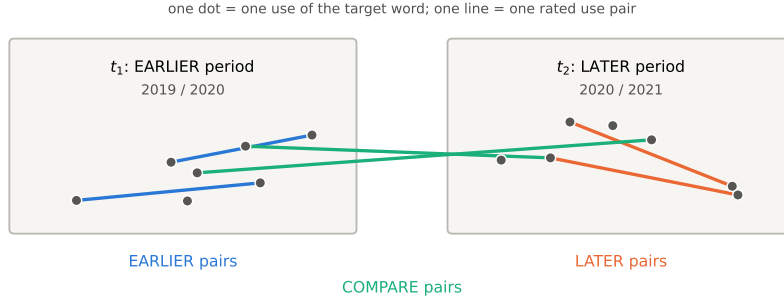


Figure 1: The three groups of use pairs. EARLIER and LATER pairs measure how homogeneous a single period is; COMPARE pairs measure the two periods against each other.

Table 3: Two TempoWiC training instances for the target word *frisk* (target in bold). The second pair contrasts the game character with the stop-and-frisk policy.

Target word: <i>frisk</i>	
2019-02	Also, just for the record, LabCorp in Frederick all but does a frisk search before your piss test.
2020-02	why did my AFRICAN dad just say that Bloomberg’s stop and frisk was good for New York? <i>gold label: same meaning</i>
2019-02	new one wut is frisk
2020-02	The stop and frisk guy is now endorsing the crime bill guy in the most shocking turn of events <i>gold label: different meaning</i>

Table 4: Alignment between Schlechtweg et al. [2018] and this report. Rows marked † are where the two studies differ by necessity rather than by choice.

Element	Schlechtweg et al. [2018]	This report
Corpus	DTA, German historical prose	TempoWiC, English tweets
Time periods	1750–1800 vs. 1850–1900	year Y vs. $Y+1$ (2019–2021)
Definition of a use	sentence, with neighbouring sentences	one tweet
Target words	22	34
Use pairs	1,320	10,097
Pairs per word and group	20	100 (EARLIER / LATER)
Groups	EARLIER, LATER, COMPARE	identical
Relatedness scale	1–4, 0 = cannot decide	1–4, mapped from model scores
Raters†	5 human annotators	5 relatedness models
Within-period pairs†	sampled from the corpus	sampled from TempoWiC’s tweets
Agreement analysis	pairwise Spearman between raters	identical, plus vs. gold labels
Change measures	Δ LATER, COMPARE, Δ COMPARE	identical
Validation of measures†	qualitative, historical dictionaries	numerical, vs. human change rate

TempoWiC. TempoWiC [Loureiro et al., 2022a] is the dataset of the EvoNLP 2022 shared task. One instance is a target word with two tweets, one per time period, and a binary label: *same meaning* or *different meaning*. The tweets are tokenised, the character span of the target is given, and every tweet carries a YYYY-MM date. Gold test labels were released in 2023. Table 3 shows two real instances for *frisk*, one of each label: both pairs share little vocabulary, and only the meaning of the target separates them.

4 Adapting DUREl to TempoWiC

4.1 How this study lines up with the original

Table 4 states the correspondence between the German study and this one, element by element. Everything in the left column is kept; only the material it is applied to changes. Three cells are genuine departures and are discussed where they matter: the annotators (§6.2), the within-period pairs, which TempoWiC does not supply and we sample (§4.3), and the validation of the measures, which we can do numerically (§6.4).

Table 5: TempoWiC splits as used here. Only gold-labelled test instances are kept; the dummy instances in the released test file are discarded.

Split	Words	COMPARE pairs	same	different	t_1/t_2
Train	15	1,428	650	778	2019/2020
Validation	4	396	172	224	2019/2020
Test	15	1,473	539	934	2020/2021
All	34	3,297	1,361	1,936	–

Table 6: Use pairs per DUREl group.

Group	Use pairs	Source
EARLIER (t_1, t_1)	3,400	sampled (100 per word)
LATER (t_2, t_2)	3,400	sampled (100 per word)
COMPARE (t_1, t_2)	3,297	original TempoWiC pairs (gold-labelled)
Total	10,097	

Table 7: Inventory per target word.

Per target word	min	median	max
Unique uses in t_1	76	98	100
Unique uses in t_2	76	98	100
COMPARE pairs (gold)	76	98	100
EARLIER / LATER pairs sampled	100	100	100

4.2 Target words and periods

We use all 34 target words from the train, validation and test splits. For every word, TempoWiC pairs a tweet from month m of year Y with a tweet from month m of year $Y+1$. We take year Y as t_1 and year $Y+1$ as t_2 (Table 5). The gap is therefore one year, against the century between DTA’s two periods. This is in line with the original recommendation to use directly adjacent and short time periods.

4.3 Rebuilding the three groups

Every TempoWiC instance is a COMPARE pair by construction. EARLIER and LATER pairs do not exist in the benchmark, so we sample them from the same tweets. For each word we collect all unique uses inside one period and draw 100 distinct pairs per word and period at random (seed 42). As in DUREl, no use pair is sampled twice within a group. Table 6 gives the result: 3,400 EARLIER pairs, 3,400 LATER pairs and 3,297 COMPARE pairs over 6,594 unique uses.

4.4 Sampling and sanity checks

Table 7 summarises what is available per word. A word has between 76 and 100 unique uses per period, which is enough for 100 distinct within-period pairs but not enough for every pair to use a distinct tweet, so a single use may appear in more than one pair. DUREl allows exactly this when the pool is small.

`src/verify.py` checks three properties of the construction, all of which hold: no use pair occurs twice inside a group, no EARLIER or LATER pair mixes the two years, and no COMPARE pair sits inside one year. Uses are deduplicated by tweet text plus the target’s character span, so one tweet is never two uses.

4.5 Models instead of annotators

DUREl’s five students are replaced by five relatedness models, each mapping a use pair to a graded relatedness score. **Jaccard** is the content-word overlap of the two tweets, a purely lexical control. **Sentence cosine** is the cosine between the mean-pooled tweet vectors. **Target cosine** is the cosine between the contextual vectors of the target word itself at encoder layer k , where the target vector is the mean over its sub-word pieces and the layer is selected on the validation split. **Similarity logistic regression** is trained on the training split over all 15 similarity features (13 layer cosines, sentence cosine, Jaccard), the same family as the strongest official shared-task baseline. The **fine-tuned cross-encoder** is trained on the same split.

Scores are mapped onto DUREl’s 1–4 scale. For the classifiers we use $\text{rel} = 1 + 3\hat{p}$, where $\hat{p}$ is the probability of *same meaning*; for the similarity scores we use min–max scaling. Both mappings are monotone, so the rankings and correlations below do not depend on them. Only the absolute means do.

Table 8: Training and scoring settings.

Setting	Value
Encoder	RoBERTa-base (12 layers, 768 dim, 125M parameters)
Maximum sequence length	128 word pieces
Optimiser	AdamW, weight decay 0.01
Learning rate	2×10^{-5} , linear decay, 10% warm-up
Batch size	16
Epochs	4 (best validation epoch kept)
Seeds	3
Hardware	2 CPU cores, no GPU
Training time	≈ 40 min per seed
Threshold (unsupervised scores)	tuned on validation for macro-F1
Logistic regression	standardised features, L_2 , $C = 1$

5 Experimental setup

Encoder. All contextual models use RoBERTa-base as shipped inside the spaCy model package `en_core_web_trf` [Honni-bal et al., 2020]. The experiments ran offline with no access to the model hub, so we extracted the encoder from that package and converted it to the standard format, checking the conversion against spaCy’s own forward pass (cosine ≥ 0.999 for every token of the test sentences). Two caveats follow: the checkpoint was trained further inside spaCy’s OntoNotes pipeline, so it is not bit-identical to the released `roberta-base`, and the same constraint rules out TimeLMs [Loureiro et al., 2022b], the time-specific Twitter encoders used by the shared-task baselines.

Supervised models. The similarity logistic regression is trained on the 1,428 training pairs with standardised features and L_2 regularisation. The cross-encoder input is `<s> tweet1 </s></s> tweet2 </s>`, with the target occurrence marked by asterisks in both tweets. Training uses AdamW, a learning rate of 2×10^{-5} , batch size 16, four epochs, a maximum length of 128 word pieces and a linear schedule with 10% warm-up, on CPU. We keep the epoch with the best validation macro-F1 and report the mean and standard deviation over 3 seeds.

Metric. Macro-F1, the official TempoWiC metric, computed on the gold-labelled COMPARE pairs of the relevant split (1,473 pairs for test). Macro-F1 is the mean of the F1 scores for *same meaning* and *different meaning*, so a system that always predicts the majority class is heavily penalised, which is why the majority baseline in Table 9 scores far below chance-level accuracy.

Compute. Everything ran on two CPU cores without a GPU. Encoding the 6,594 unique uses takes about 11 minutes, fine-tuning about 40 minutes per seed, and scoring all 10,097 pairs with the cross-encoder about 19 minutes. The full pipeline, including the three seeds, takes roughly three hours. Table 8 lists the settings.

6 Results

6.1 Pairwise relatedness

Table 9 gives the results on the TempoWiC test set. Lexical overlap is barely above chance, at 53.5 macro-F1 against 50.6 for random scores, so the benchmark cannot be solved by surface similarity. Zero-shot cosine between target-word vectors is much stronger and depends heavily on the layer (Figure 2). Sense information peaks in the middle layers on the validation words, while the optimum on the test words sits higher. The layer picked on the validation split, layer 5, therefore transfers to 63.6 macro-F1, where an oracle layer 8 would give 70.3. With only four validation words, layer selection is unstable.

Supervision does not help here. The similarity logistic regression reaches 65.4 and the fine-tuned cross-encoder 61.9 ± 1.4 over 3 seeds. Both are below plain last-layer cosine (68.0). The reason is the split design: the training split has 15 target words and the test split 15 different ones, so a supervised model can fit cues that are specific to the training words, such as which sense of *turnip* or *villager* is frequent, and those cues do not transfer. The official shared-task baselines show the same ordering. Logistic regression over TimeLMs similarities reaches 70.3 macro-F1, fine-tuned RoBERTa-large reaches 59.1 and the winning system 77.1 [Lyu et al., 2022]. Our fine-tuned base-size model, trained on CPU and without time-specific pre-training, sits inside that range.

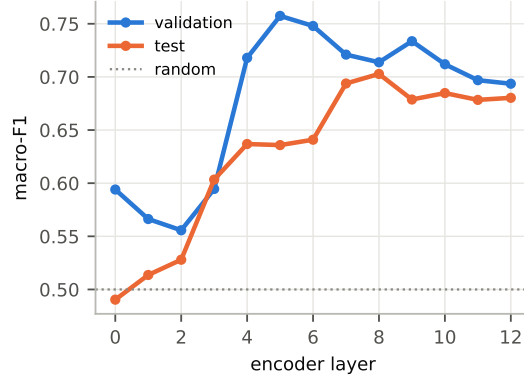
6.2 Agreement between relatedness models

DUREl’s Table 3 reports pairwise Spearman correlations between annotators, and each annotator against the average of the others. Table 10 is the same table with models in place of annotators, computed on the gold-labelled COMPARE pairs. Three things stand out.

First, the unsupervised contextual models agree with each other much more than they agree with people: up to $\rho = 0.84$ between adjacent layers, against $\rho \leq 0.36$ with the human labels. The German study shows the opposite pattern, with an

Table 9: TempoWiC test results (%). Thresholds for the unsupervised scores are tuned on the validation split.

System	macro-F1	F1 _{same}	F1 _{diff}	Acc.
<i>Controls</i>				
Random	50.6	41.5	59.7	52.3
Majority class	38.8	0.0	77.6	63.4
Lexical overlap (Jaccard)	53.5	39.1	67.9	58.0
<i>Zero-shot contextual similarity</i>				
Sentence vectors (mean-pooled)	55.1	50.8	59.3	55.5
Target vectors, last layer (L12)	68.0	59.3	76.7	70.4
Target vectors, layer selected on dev (L5)	63.6	52.3	74.9	67.1
(oracle layer L8)	70.3	62.3	78.3	72.4
<i>Supervised</i>				
Logistic regression over similarities	65.4	53.3	77.5	69.7
Fine-tuned cross-encoder (seed 13)	59.9	40.4	79.3	69.3
Fine-tuned cross-encoder (seed 42)	63.1	46.8	79.4	70.3
Fine-tuned cross-encoder (seed 7)	62.6	45.5	79.8	70.5
Fine-tuned cross-encoder (mean±sd)	61.9±1.4	44.2±2.8	79.5±0.2	70.1±0.5

**Figure 2:** Zero-shot macro-F1 by encoder layer. Sense information peaks in the middle layers. The last layers are shaped by the encoder’s own training objective.**Table 10:** Spearman correlations between relatedness models and the human labels on COMPARE pairs (DUREl Table 3 analogue). The last row gives each model against the average of the other models.

	cos L5	cos L8	cos L12	sent. cos	Jaccard	sim. log. reg.	fine-tuned CE	GOLD
cos L5	–	0.76	0.61	0.16	0.19	0.64	0.19	0.33
cos L8		–	0.84	0.21	0.10	0.62	0.24	0.36
cos L12			–	0.27	–0.00	0.44	0.29	0.30
sent. cos				–	–0.25	0.49	0.24	0.18
Jaccard					–	–0.13	–0.18	–0.02
sim. log. reg.						–	0.29	0.39
fine-tuned CE							–	0.58
GOLD								–
vs. avg. of others	0.63	0.71	0.64	0.36	–0.09	0.61	0.31	–

average pairwise ρ of 0.66 between its five annotators. A large part of the agreement between our models comes from shared architecture, not from shared understanding of the words.

Second, supervision shifts the picture in one direction only. The fine-tuned cross-encoder is by far the closest to the human labels ($\rho = 0.58$) but correlates only weakly with the unsupervised scores (0.19 to 0.29): it has learnt a decision boundary rather than a relatedness ordering, which is useful for classification and less useful for graded annotation. The similarity regression sits in between ($\rho = 0.39$ with gold, 0.62 to 0.64 with the layer cosines) and is the better choice when the analysis needs graded scores. Third, the lexical control is uncorrelated with the gold labels (–0.02), the check that the relatedness signal really is contextual.

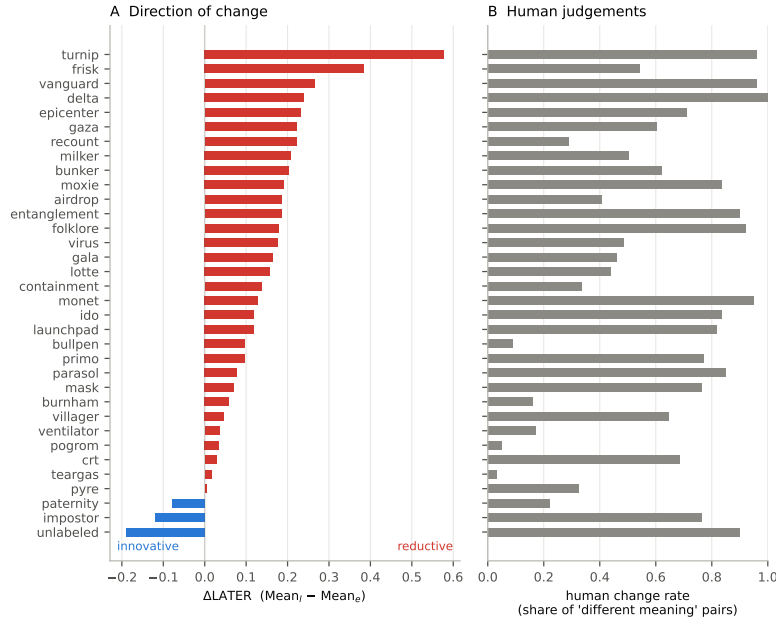


Figure 3: Target words ranked by ΔLATER (panel A) next to the human change rate (panel B). The framework reads a positive ΔLATER as reductive change. Here it mostly reflects pandemic-driven concentration of usage in the later period.

6.3 Word-level change

Figure 3 ranks all 34 target words by ΔLATER next to the human change rate, that is, the share of a word’s COMPARE pairs labelled *different meaning*. The German study found a clear three-way split into reductive (> 0), innovative (< 0) and stable (≈ 0) words. Our distribution is strongly asymmetric: 31 of 34 words have a positive ΔLATER . Read literally, that would mean almost every target word lost a meaning between 2019 and 2021, which is not plausible.

The cause is the corpus effect that Schlechtweg et al. [2018] warn about. TempoWiC’s later period is 2020–2021, and the target words include *mask*, *virus*, *delta*, *containment*, *epicenter* and *ventilator*. In those years the words are used in one pandemic-related way most of the time, so later uses resemble each other more than earlier ones did and Mean_l goes up. DUREL reads that extra within-period homogeneity as meaning loss, when what changed is topic prevalence. This is the main lesson of the transfer: ΔLATER assumes a corpus that is balanced across periods, and a Twitter sample spanning 2020 is not.

6.4 Do the measures track human judgements?

The German study could only validate its measures qualitatively, against what the historical literature says about each word. TempoWiC lets us do it with numbers, because each word’s human change rate is an independent measurement of how much it changed. Table 11 correlates that rate with each DUREL measure.

COMPARE behaves as the framework predicts. A low COMPARE value means strong change, and the correlation is $\rho = -0.61$ ($p = 0.0001$) for the model our selection protocol picks, namely cos L5 with the best validation macro-F1 (Figure 4, left). This does not depend on that choice. Every contextual model lands between -0.35 and -0.68 , the supervised similarity model is the strongest at $\rho = -0.68$, and the lexical control shows no relationship at all.

ΔLATER is only weakly related to the human change rate ($\rho = 0.34$) and is insignificant for several models, which matches §6.3. The polysemy-normalised $\Delta\text{COMPARE}$, proposed in the original paper as a refinement, does not improve on plain COMPARE here ($\rho = -0.23$). Subtracting Mean_e removes real signal, because in a one-year Twitter window the earlier-period spread of a word is driven by topical variety rather than by stable polysemy.

6.5 Qualitative analysis

Figure 5 contrasts the most stable and the most changed target word, in the same form as DUREL’s Figure 4: the distribution of relatedness judgements per group. For *teargas* (human change rate 0.03) the three distributions are almost identical, with group means of 2.89, 2.91 and 2.87. The word is used the same way in both periods and across them. For *delta* (change rate 1.00, with the Greek letter and the airline giving way to the virus variant) the COMPARE distribution shifts towards low relatedness, $\text{Mean}_c = 2.53$, while the LATER distribution is the most concentrated at high relatedness, $\text{Mean}_l = 2.78$, because the 2021 uses are nearly all the new sense. That is the pattern the framework is built to show. It also explains why ΔLATER labels this word reductive (+0.24) when a new dominant meaning is what actually arrived.

Table 11: Spearman correlation of each DUREl measure with the human change rate across 34 words. $*p < 0.05$, $**p < 0.01$. COMPARE is expected to correlate negatively.

Relatedness model	COMPARE	Δ COMPARE	Δ LATER
cos L5	-0.61**	-0.23	0.34*
cos L8	-0.62**	-0.23	0.22
cos L12	-0.41*	-0.19	0.14
sim. log. reg.	-0.68**	-0.17	0.26
fine-tuned CE	-0.64**	-0.21	-0.24
sent. cos	-0.35*	-0.24	-0.06
Jaccard	0.15	0.04	0.36*

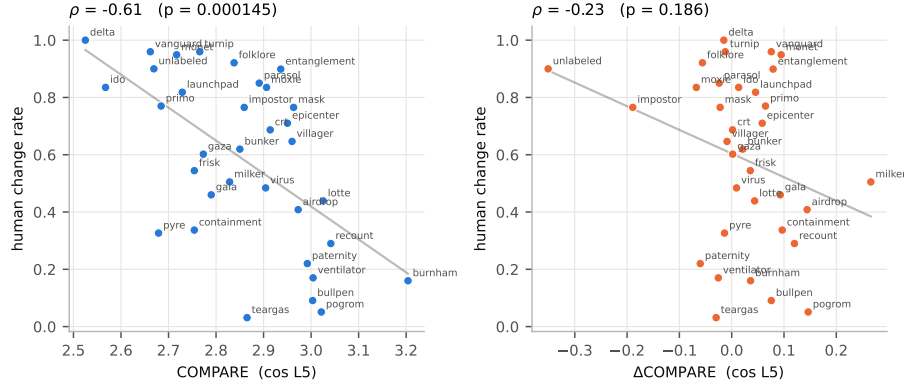


Figure 4: Word-level measures against the human change rate.

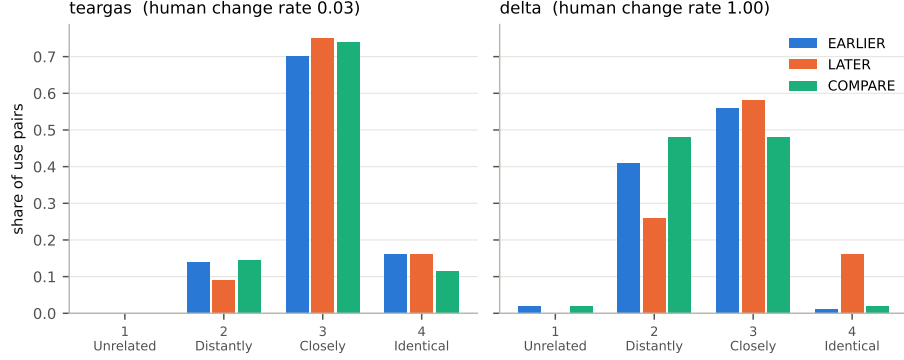


Figure 5: Relatedness distributions per group (DUREl Figure 4 analogue) for the most stable and the most changed target word.

6.6 Where the models fail

Table 12 breaks test accuracy down by target word and compares it against the per-word majority baseline, that is, always predicting the more frequent label for that word. The picture is sobering: the dev-selected cosine falls below that baseline on 11 of the 15 test words, and per-word accuracy is correlated with how skewed the word’s labels are ($\rho = 0.60$, $p = 0.017$). Words whose pairs are nearly all *different meaning* (delta, 86.7%) or nearly all *same meaning* are easy; words with a balanced mix (bullpen, 42.4%) are hard.

The fine-tuned cross-encoder changes the shape of the errors rather than their number. It is better on the strongly shifted words and much worse on the stable ones, because the training words pushed it towards predicting *different meaning*. The word *burnham* is the clearest case: its pairs are *same meaning* 84% of the time, the zero-shot model scores 75.0% on it, and the fine-tuned model 21.0%.

This matters for the DUREl measures. Per-pair accuracy is uneven across words, but the word-level means average over roughly 100 pairs, and COMPARE still correlates at $\rho = -0.61$ with the human change rate. A measure built on group means is more robust than the per-pair judgements it is built from, which is the practical argument for keeping DUREl’s design when the raters are models.

Table 12: Test accuracy per target word (%) against the per-word majority baseline (maj.), sorted by the accuracy of the dev-selected cosine. Every word has 91–100 test pairs. chg.: human change rate; FT CE: fine-tuned cross-encoder.

Word	chg.	maj.	cos L5	FT CE
delta	1.00	100.0	86.7	92.9
unlabeled	0.90	90.0	83.0	81.0
ido	0.84	83.5	82.4	83.5
burnham	0.16	84.0	75.0	21.0
monet	0.95	94.9	73.5	89.8
moxie	0.84	83.5	71.1	88.7
vanguard	0.96	96.0	69.7	91.9
launchpad	0.82	81.8	69.7	83.8

Word	chg.	maj.	cos L5	FT CE
crt	0.69	68.7	68.7	68.7
gaza	0.60	60.2	67.3	62.2
gala	0.46	54.0	61.0	48.0
milker	0.51	50.5	54.5	62.6
airdrop	0.41	59.2	54.1	39.8
pyre	0.33	67.3	48.0	62.2
bullpen	0.09	90.9	42.4	80.8

7 Discussion

What transfers. Use pairs, the three groups and the relatedness scale need no language-specific machinery, and COMPARE remains a valid measure of the degree of change, now with numerical support the original study could not provide. The claim that DUREl is language-independent holds, and extends to genre: the framework works on 280-character tweets as well as on 19th-century prose.

What does not. Δ LATER is fragile. It infers change from the difference between two within-period means, so any process that changes how topically concentrated a period is will move it, whether or not a meaning was gained or lost. A pandemic, an election or a product launch all do this. The German study meets the same problem in mild form as genre drift in DTA; on social media it dominates. The original recommendation, a corpus balanced across periods, is essentially unobtainable for Twitter data spanning 2020.

Models as annotators. Using models instead of annotators is what makes this scale: 10,097 pairs and no annotation budget. The substitution is lossy, since models agree with each other far more than with people, and a benchmark whose splits share no target words additionally penalises any model that has learnt word-specific associations. At the word level this matters less than one might expect, because the errors partly average out over 100 pairs per word and COMPARE still correlates at $\rho = -0.61$. Per-pair judgements from a model, however, should not be presented as a test set, and a relatedness model tuned on a handful of words cannot be assumed to give calibrated relatedness on new ones.

A checklist for modern corpora. Report COMPARE as the primary measure and treat Δ LATER as a diagnostic, not as evidence of a change type. Check the corpus for events that concentrate usage before reading anything into Δ LATER; plotting Mean_e and Mean_l separately, as in Table 13, exposes such an effect at once. And if models do the rating, prefer one that produces graded scores over one fine-tuned for a binary decision, since the measures are means and a saturated probability carries no gradation.

Limitations. No time-specific encoder was available offline, so the absolute task numbers sit below published systems; EARLIER and LATER pairs have no human labels, so only COMPARE means can be checked against people; 34 words is more than DUREl’s 22 but still small for correlation analysis; and CPU-only training limited model size and the hyper-parameter search.

8 Conclusion

We moved DUREl from German historical prose to English social media by rebuilding its three-group design on top of TempoWiC and replacing human annotators with relatedness models. COMPARE survives the transfer and is validated numerically against independent human judgements of change ($\rho = -0.61$). Δ LATER does not survive it: on a corpus spanning 2020 it mostly measures how far a word’s usage narrowed onto one topic. For anyone applying DUREl to contemporary corpora, the practical recommendation is to report COMPARE as the primary measure and to treat Δ LATER as a diagnostic that is only reliable when the corpus is balanced across periods.

Three extensions follow naturally. The obvious one is to run the same pipeline with a time-specific encoder, since TimeLMs is exactly the kind of model the framework would benefit from and was only unavailable here for environmental reasons. The second is to replace group means with the Word Usage Graph machinery of Schlechtweg et al. [2021], which clusters uses instead of averaging over pairs and would be less sensitive to the topic-concentration effect of §6.3. The third is a small human annotation of EARLIER and LATER pairs, which would turn the model-based within-period means into measured ones and make the whole three-group design checkable against people rather than only its COMPARE part.

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A Per-word measures

Table 13: Per-word DUREl measures. Sp.: TempoWiC split (tr/va/te). chg. rate: share of COMPARE pairs labelled *different meaning* by human annotators.

Word	Sp.	chg.	Mean _e	Mean _t	Mean _c	ΔL	ΔC
delta	te	1.00	2.54	2.78	2.53	+0.24	-0.02
ido	te	0.84	2.55	2.67	2.57	+0.12	+0.01
vanguard	te	0.96	2.59	2.85	2.66	+0.27	+0.08
unlabeled	te	0.90	3.02	2.83	2.67	-0.19	-0.35
pyre	te	0.33	2.69	2.70	2.68	+0.00	-0.01
primo	va	0.77	2.62	2.72	2.68	+0.10	+0.06
monet	te	0.95	2.62	2.75	2.72	+0.13	+0.09
launchpad	te	0.82	2.68	2.80	2.73	+0.12	+0.05
containment	tr	0.34	2.66	2.80	2.75	+0.14	+0.10
frisk	tr	0.54	2.72	3.10	2.75	+0.38	+0.04
turnip	tr	0.96	2.78	3.36	2.77	+0.58	-0.01
gaza	te	0.60	2.77	2.99	2.77	+0.22	+0.00
gala	te	0.46	2.70	2.86	2.79	+0.16	+0.09
milkier	te	0.51	2.56	2.77	2.83	+0.21	+0.27
folklore	tr	0.92	2.89	3.07	2.84	+0.18	-0.06
bunker	tr	0.62	2.83	3.03	2.85	+0.20	+0.02
impostor	va	0.77	3.05	2.93	2.86	-0.12	-0.19

Word	Sp.	chg.	Mean _e	Mean _t	Mean _c	ΔL	ΔC
teargas	tr	0.03	2.89	2.91	2.87	+0.02	-0.03
parasol	tr	0.85	2.91	2.99	2.89	+0.08	-0.02
virus	tr	0.48	2.90	3.07	2.90	+0.18	+0.01
moxie	te	0.84	2.97	3.17	2.91	+0.19	-0.07
crt	te	0.69	2.91	2.94	2.91	+0.03	+0.00
entanglement	tr	0.90	2.86	3.04	2.94	+0.19	+0.08
epicenter	tr	0.71	2.89	3.12	2.95	+0.23	+0.06
villager	tr	0.65	2.97	3.02	2.96	+0.05	-0.01
mask	tr	0.77	2.99	3.06	2.96	+0.07	-0.02
airdrop	te	0.41	2.83	3.01	2.97	+0.19	+0.14
paternity	tr	0.22	3.05	2.97	2.99	-0.08	-0.06
bullpen	te	0.09	2.93	3.03	3.00	+0.10	+0.08
ventilator	tr	0.17	3.03	3.07	3.00	+0.04	-0.03
pogrom	tr	0.05	2.88	2.91	3.02	+0.03	+0.15
lotte	va	0.44	2.98	3.14	3.03	+0.16	+0.04
recount	va	0.29	2.92	3.14	3.04	+0.22	+0.12
burnham	te	0.16	3.17	3.23	3.20	+0.06	+0.04

B Reproduction

Pipeline. The scripts run in this order, each writing into `results/`: `data_prep.py` (builds the three groups), `extract_encoder.py` (converts the encoder), `encode.py` (target vectors at 13 layers), `zeroshot.py` and `simlr.py` (unsupervised scores and the similarity regression), `finetune.py` and `score_all.py` (cross-encoder), `durel.py` (measures and agreement), `figures.py` and `make_tables.py`, and `verify.py`.

Verification and provenance. `verify.py` recomputes the headline numbers with a second implementation: macro-F1 by hand rather than with `scikit-learn`, the group means from `pairs.jsonl` rather than from the pipeline’s data frames, and the structural properties of §4.3 re-checked. It also compares the numbers quoted in this report against the result files; all 42 checks pass. Every number here is generated into `results/` and `report/tables/` and read in through `\input.use_vectors.npz` (261 MB) is left out of the archive and is regenerated by `extract_encoder.py` and `encode.py`, which need the `spaCy` package `en_core_web_trf` 3.7.3.